

# UNIRATIONALITY OF EVERY SMOOTH $(2, 3)$ HYPERSURFACE IN $\mathbf{P}^2 \times \mathbf{P}^2$ BY TANGENT RESIDUALS AND LATTÈS VARIATION: THE CONSTRUCTION AND ITS VERIFICATION ARTIFACTS

ABSTRACT. Let  $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$  be any smooth hypersurface of bidegree  $(2, 3)$  over  $\mathbf{C}$ . We prove that  $X$  is unirational. A suitable vertical surface  $S = X \cap (\mathbf{P}^2 \times L)$  is rational by Tsen's theorem and maps birationally, by fiberwise tangent residuals in the plane-cubic fibration, onto a horizontal divisor  $T \sim aH_x + H_y$  with  $1 \leq a \leq 10$ . This gives a dominant rational parametrization of even degree  $2a \leq 20$ . The key uniform input is the quartic residual-line Lattès map of a smooth plane cubic: its critical-value curve is the dual sextic, so it recovers the embedded cubic and cannot remain constant in a smooth nonconstant family. For a general equation a universal resultant identity gives  $T \sim 10H_x + H_y$  and degree exactly 20. Combined with known results, the construction yields smooth threefolds that are unirational and irrational, with very general members additionally not stably rational. These notes are self-contained on the mathematics and additionally document every machine computation the proof relies on, with pointers to the certificate files in the accompanying repository.

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## 1. INTRODUCTION AND STATEMENT

A *standard conic bundle* over  $\mathbf{P}^2$  is a smooth projective threefold  $X$  with a flat morphism  $\pi: X \rightarrow \mathbf{P}^2$  whose fibers are plane conics, with relative Picard rank one. Its discriminant  $\Delta \subset \mathbf{P}^2$  is a curve; the birational geometry of  $X$  is largely governed by  $\deg \Delta$ . Rationality is classical:  $X$  is rational iff  $\deg \Delta \leq 4$ , or  $\deg \Delta = 5$  with even theta-characteristic (Beauville, Shokurov; see [14] for a survey), while a conic bundle with  $\deg \Delta \geq 6$  is irrational by the Chow/Prym criterion [2, Théorème 4.9]. *Unirationality* is far more delicate. The sharpest published statement is [10, Corollary 4.9]: over an algebraically closed field, a smooth conic bundle  $Q^1 \rightarrow \mathbf{P}^2$  whose discriminant carries a point of multiplicity  $m_p$  with

$$\delta \leq m_p + 7$$

is unirational. The mechanism is projection from that point, which drops the residual discriminant degree by  $m_p$ ; consequently a *smooth* discriminant ( $m_p = 1$ ) is covered only up to  $\delta \leq 8$ , and degree 9 only when  $\Delta$  is singular. The unpublished preprint [12] reaches the same boundary in the same shape. (The frequently quoted bound  $\delta \leq 7$  of [9] concerns conic bundle *surfaces*, where  $(-K)^2 = 8 - \delta$ , not threefolds over  $\mathbf{P}^2$ .) For large general discriminants the expectation in the literature was negative: “*It is likely that a conic bundle with sufficiently large and general discriminant curve is not unirational*” [14, §14.3], echoed in [11].

Every smooth hypersurface  $X$  of bidegree  $(2, 3)$  in  $\mathbf{P}^2 \times \mathbf{P}^2$  is a standard conic bundle over the second factor with discriminant of degree 9; the general discriminant is smooth. It therefore misses the published boundary by exactly one unit. These notes give a complete proof of:

**Theorem 1.1.** *Every smooth hypersurface  $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$  of bidegree  $(2, 3)$  over  $\mathbf{C}$  is unirational. More precisely, there is a dominant rational map  $\mathbf{P}^3 \dashrightarrow X$  of even degree at most 20.*

**Theorem 1.2** (Generic degree). *For a general  $(2, 3)$  equation, Construction 3.1 gives a dominant rational map  $\mathbf{P}^3 \dashrightarrow X$  of degree exactly 20.*

**Corollary 1.3.** *Every smooth  $(2, 3)$  hypersurface is unirational (Theorem 1.1) and irrational [2, Théorème 4.9]. A very general member is in addition not stably rational [3, Theorem 1.1]; for such a member,  $\mathrm{Br}(X) = 0$  and  $H_{\mathrm{nr}}^3(X, \mathbf{Q}/\mathbf{Z}(2)) = 0$  [5], so the failure of stable rationality is invisible to the classical unramified invariants; unlike the Artin–Mumford threefold [1], whose non-stable-rationality is detected by  $\mathrm{Br}$ , these examples are detected only by degeneration of the diagonal [3].*

The construction is elementary in style and could have been carried out classically; the nearest antecedent is the 1980 note of Del Centina–Verdi [6], which asserted a *special* smooth unirational  $(3, 2)$  example by an entirely different method (rationality of one double plane with special branch sextic). Their displayed data contain inconsistencies; a corrected pencil is certified in the repository (§5). The fiberwise tangent-residual mechanism used here appears to be new.

**Provenance of these notes.** The construction and proof below were developed in a computational project archived at <https://github.com/mattrobball/unirational>. Its finite algebraic identities were mechanically verified. The mathematical exposition is self-contained; §5 is a complete ledger of the machine artifacts the proof relies on, what each verifies, and which steps rest on standard prose mathematics instead.

## 2. SETUP

**Convention 2.1.** We work over  $\mathbf{C}$  (any algebraically closed field of characteristic 0 works verbatim). Coordinates  $x = (x_0:x_1:x_2)$  on  $\mathbf{P}_x^2$  and  $y = (U:V:W)$  on  $\mathbf{P}_y^2$ . A bidegree-(2, 3) form is written

$$F(x, y) = x^\top A(y) x,$$

with  $A(y)$  a symmetric  $3 \times 3$  matrix of cubic forms in  $y$ . We write  $X = \{F = 0\}$  and let  $H_x, H_y$  denote the restrictions of the two hyperplane classes. Unless a generic refinement is explicitly stated,  $X$  is assumed smooth. “General” means: in a dense Zariski-open subset of the parameter space  $\mathbf{P}^{59} = \mathbf{P}H^0(\mathcal{O}_{\mathbf{P}^2 \times \mathbf{P}^2}(2, 3))$ , intersected with the smooth locus.

The threefold  $X$  carries *two* fibrations:

- $\pi = \text{pr}_y: X \rightarrow \mathbf{P}_y^2$ , a flat conic bundle with discriminant  $\Delta = \{\det A(y) = 0\}$  of degree 9 (smooth for general  $F$ );
- $\rho = \text{pr}_x: X \rightarrow \mathbf{P}_x^2$ , a fibration whose fiber over  $x$  is the plane cubic  $C_x = \{F(x, -) = 0\} \subset \mathbf{P}_y^2$ ; its generic fiber is smooth by generic smoothness.

The interplay of the two is the whole story.

**Lemma 2.2.** *For smooth  $X$ :  $\text{Pic}(X) = \mathbf{Z}H_x \oplus \mathbf{Z}H_y$  and  $K_X = \mathcal{O}_X(-1, 0)$ . Every effective divisor class  $aH_x + bH_y$  has  $a, b \geq 0$ . Writing  $f_y$  (a conic fiber of  $\pi$ ) and  $c_x$  (a cubic fiber of  $\rho$ ) for the two curve classes,*

$$H_x \cdot f_y = 2, \quad H_y \cdot f_y = 0, \quad H_x \cdot c_x = 0, \quad H_y \cdot c_x = 3.$$

*Consequently a surface  $S \in |aH_x + bH_y|$  has degree  $2a$  over  $\mathbf{P}_y^2$  and degree  $3b$  over  $\mathbf{P}_x^2$ . In particular every multisection of the conic bundle has even degree, and  $\rho$  has no rational section.*

*Proof.* Grothendieck–Lefschetz for the ample divisor  $X$  in  $\mathbf{P}^2 \times \mathbf{P}^2$  ( $\dim X = 3$ ) gives  $\text{Pic}$ ; adjunction gives  $K_X$ . Effectivity: from  $0 \rightarrow \mathcal{O}(a-2, b-3) \rightarrow \mathcal{O}(a, b) \rightarrow \mathcal{O}_X(a, b) \rightarrow 0$  and  $H^1(\mathbf{P}^2 \times \mathbf{P}^2, \mathcal{O}(m, n)) = 0$  for all  $m, n$ , one gets  $h^0(\mathcal{O}_X(a, b)) = h^0(\mathcal{O}(a, b)) - h^0(\mathcal{O}(a-2, b-3))$ , nonzero only if  $a, b \geq 0$ . The intersection numbers are immediate. The parity statement is consistent with, and geometrically explains, the nontriviality of the 2-torsion Brauer class of the generic conic (restriction–corestriction).  $\square$

*Remark 2.3.* Since every surface in the threefold  $X$  is a divisor, Lemma 2.2 constrains all of them. Adjunction gives  $K_S = ((a-1)H_x + bH_y)|_S$  for  $S \in |aH_x + bH_y|$ ; for a *horizontal* class ( $a \geq 1$ ) this is effective or trivial, so *no smooth member of any horizontal linear system is rational*. Any horizontal surface with rational normalization — which Theorem 1.1 requires, by Proposition 2.4 below — is necessarily a singular member of its system. This is the precise sense in which the problem resists naive searches, and why the construction below must produce a highly singular divisor.

**Proposition 2.4** (Multisection principle). *Let  $T \subset X$  be an irreducible surface dominating  $\mathbf{P}_y^2$  with rational normalization  $\tilde{T}$ , of degree  $N$  over  $\mathbf{P}_y^2$ . Then  $X$  is unirational of degree  $N$ . Conversely, if  $X$  is unirational it contains such surfaces.*

*Proof.* Base-change:  $X \times_{\mathbf{P}_y^2} \tilde{T} \rightarrow \tilde{T}$  is a conic bundle over a rational surface carrying the tautological section  $t \mapsto (t, t)$ . A conic with a rational point over the field  $\mathbf{C}(\tilde{T})$  is isomorphic to  $\mathbf{P}^1$ , so  $X \times_{\mathbf{P}_y^2} \tilde{T}$  is birational to  $\tilde{T} \times \mathbf{P}^1$ , a rational threefold; the projection to  $X$  is dominant of degree  $N$ . For the converse, the image of a general plane under a dominant map

$\mathbf{P}^3 \dashrightarrow X$  is a horizontal surface with rational (indeed unirational, hence by Castelnuovo rational) normalization.  $\square$

Harris's formulation of the problem — find rational surfaces in  $X$  *other than* the preimages of rational curves in  $\mathbf{P}_y^2$  [8, p. 24] — asks exactly for the horizontality in Proposition 2.4: the preimage  $\pi^{-1}(\text{curve})$  is vertical and useless. The point of the construction below is that one of these “useless” vertical surfaces can be *converted* into a horizontal one using the second fibration  $\rho$ .

### 3. THE GENERIC DEGREE-20 CONSTRUCTION

In this section  $F$  is general. The universal identity survives every specialization; Section 4 will use its full line-space variation to remove this generality assumption.

**Construction 3.1.** Fix a general line  $L \subset \mathbf{P}_y^2$  and choose coordinates with  $L = \{W = 0\}$ .

- (I) **The vertical seed.** Let  $S = X \cap (\mathbf{P}_x^2 \times L) \in |\mathcal{O}_X(0, 1)|$ . Projection to  $L \simeq \mathbf{P}^1$  makes  $S$  a conic bundle over  $\mathbf{P}^1$ ; its generic fiber is a smooth conic over the  $C_1$  field  $\mathbf{C}(L) \simeq \mathbf{C}(t)$ , hence has a rational point by Tsen's theorem, and  $S$  is a *rational* surface (irreducible, since the generic fiber is geometrically integral). Under the other projection,  $S$  is a *trisection* of  $\rho$ : for general  $x$ , the fiber is  $C_x \cap L = \{p_1, p_2, p_3\}$ .
- (II) **Tangent residuals.** For general  $x$  the cubic  $C_x$  is smooth at each  $p_i$  (this needs only that the binary cubic  $g_x = F(x, -)|_L$  has distinct roots). Define

$$\tau: S \dashrightarrow X, \quad (x, p) \mapsto (x, \tau_x(p)), \quad T_p C_x \cdot C_x = 2p + \tau_x(p).$$

In the group law of  $C_x$  (any flex as origin) this is  $p \mapsto -2p$ .

- (III) **The horizontal image.** Let  $T = \tau(S)$ . Theorem 3.4 below shows  $T \in |10H_x + H_y|$  and  $\tau: S \dashrightarrow T$  is birational. By Lemma 2.2,  $T$  has degree  $T \cdot f_y = 20$  over  $\mathbf{P}_y^2$ ; it is horizontal with rational normalization, so Proposition 2.4 gives a dominant degree-20 map  $\tilde{T} \times \mathbf{P}^1 \dashrightarrow X$ , proving Theorem 1.2.

Everything therefore rests on identifying the image divisor exactly. This is done by a universal polynomial identity.

**3.1. The universal resultant identity.** Write a ternary cubic in the coordinates adapted to  $L = \{W = 0\}$  as

$$(1) \quad f(U, V, W) = g(U, V) + W h(U, V) + W^2(iU + jV) + kW^3,$$

with  $g$  a binary cubic and  $h$  a binary quadratic. For  $p = [u:v:0]$  with  $g(u, v) = 0$ , the tangent line to  $\{f = 0\}$  at  $p$ , evaluated at  $[U:V:W]$ , is

$$P_{U,V,W}(u, v) = U g_u(u, v) + V g_v(u, v) + W h(u, v),$$

because  $f_U, f_V, f_W$  restrict on  $W = 0$  to  $g_u, g_v, h$ . Set

$$\mathcal{R}_f(U, V, W) = \text{Res}_{u,v}(g, P_{U,V,W}),$$

the resultant of binary forms of degrees 3 and 2; up to the standard normalization,  $\mathcal{R}_f$  is the product of the three tangent lines at the points of  $\{g = 0\}$ , a cubic in  $(U, V, W)$ .

**Theorem 3.2** (Universal identity). *Let  $\Delta(g)$  denote the classical discriminant of the binary cubic  $(18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2)$  for  $g = au^3 + bu^2v + cuv^2 + dv^3$ . Then, identically in the ten coefficients of  $f$ ,*

$$(2) \quad \mathcal{R}_f + \Delta(g) f = W^2 q_f,$$

where  $q_f$  is a linear form in  $(U, V, W)$  whose three coefficients are homogeneous of degree 5 in the coefficients of  $f$  and have greatest common divisor 1 in the universal coefficient ring.

*Proof.* Expand the left side in  $W$ . The  $W^0$  term is  $\text{Res}(g, Ug_u + Vg_v) + \Delta(g)g(U, V)$ . By Euler's relation  $ug_u + vg_v = 3g$ , each factor  $Ug_u(p_i) + Vg_v(p_i)$  of the resultant is a linear form vanishing at the root  $p_i$ , so the product is proportional to  $g(U, V)$ ; the constant is  $-\Delta(g)$  (a classical covariant identity, verified symbolically in the artifacts). The  $W^1$  coefficient of  $\mathcal{R}_f$  is a binary quadratic in  $(U, V)$ ; evaluating at a root  $p_k$  of  $g$  kills every term of the root-expansion except the one attached to  $p_k$ , and comparison at the three roots identifies it with  $-\Delta(g)h(U, V)$  (two binary quadratics agreeing at three distinct points are equal; the case  $\Delta(g) = 0$  follows by continuity of a polynomial identity). Hence the left side of (2) is divisible by  $W^2$ , and being cubic, the quotient  $q_f$  is linear. The coefficient degrees:  $\text{Res}_{3,2}$  has bidegree (2, 3) in the coefficients of its two arguments, and the coefficients of  $P$  are linear in those of  $f$ , giving degree  $2 + 3 = 5$ ; likewise  $\deg \Delta(g) \cdot f = 4 + 1$ . Primitivity ( $\gcd = 1$ ) is a machine fact (§5, artifact A1/A4).  $\square$

*Remark 3.3* (Geometric content). Restricted to the cubic  $C = \{f = 0\}$ , the product of the three tangent lines has divisor  $\sum_i (2p_i + r_i)$  where  $r_i = \tau(p_i)$ , while  $W$  has divisor  $p_1 + p_2 + p_3$ . On the cubic,  $2p_i + r_i \sim H_y|_C$  and  $\sum p_i \sim H_y|_C$ , whence

$$r_1 + r_2 + r_3 \sim 3H_y|_C - 2H_y|_C = H_y|_C.$$

A degree-3 effective divisor on a smooth plane cubic lying in  $|H_y|_C|$  is a line section: *the three tangent residuals are collinear*, and  $\{q_f = 0\}$  is the line through them. (Group-law formulation: the residual map is  $p \mapsto -2p$  and  $(-2p_1) + (-2p_2) + (-2p_3) = -2(p_1 + p_2 + p_3) = 0$ .)

**3.2. The image divisor and its class.** Now let the coefficients of  $f = F(x, -)$  vary: they are *quadratic* forms in  $x$ . Then  $\Delta(g_x)$  has bidegree (8, 0),  $\mathcal{R}$  has bidegree (10, 3), and the covariant line  $q = q_{F(x, -)}(y)$  has bidegree (10, 1) — linear in  $y$ , with three coefficient forms  $q_U, q_V, q_W \in \mathbb{C}[x]_{10}$ .

**Theorem 3.4.** *For general  $F$ :*

- (1) *On  $X$  the identity (2) reads  $\mathcal{R} = W^2q$ , giving the exact divisor equality*  
 $\text{div}_X(\mathcal{R}) = 2S + T_q, \quad T_q := \text{div}_X(q) \in |10H_x + H_y|, \quad (10, 3) = 2(0, 1) + (10, 1).$
- (2)  *$T_q$  has no component vertical over  $\mathbf{P}_x^2$ ; its fiber over a general  $x$  is exactly the three distinct tangent residuals  $\{r_1, r_2, r_3\}$ , reduced.*
- (3)  *$T_q = T = \overline{\tau(S)}$  is irreducible and  $\tau: S \dashrightarrow T$  is birational.*

*Proof.* (1)  $q|_X \not\equiv 0$  since bidegree (10, 1) admits no factor of bidegree (2, 3), and then  $\mathcal{R}|_X = W^2q|_X \not\equiv 0$ ;  $W$  cuts  $S$  (reduced and irreducible for general  $F, L$ ).

(2) A component of  $T_q$  vertical over  $\mathbf{P}_x^2$  (i.e. of class  $(a, 0)$ ) would force  $q(x, -) \equiv 0$  along a curve of  $x$ 's, i.e. a common factor of  $q_U, q_V, q_W$ . For each potential common-factor degree  $e \geq 1$ , the locus of coefficient triples with such a factor is the image of the proper multiplication map  $\mathbf{P}(\mathbb{C}[x]_e) \times \mathbf{P}(\mathbb{C}[x]_{10-e})^2 \rightarrow \mathbf{P}(\mathbb{C}[x]_{10})^3$ -incidence, hence closed; there are finitely many  $e$ ; so  $\{F : \gcd(q_U, q_V, q_W) = 1\}$  is open. It is nonempty by an explicit witness (artifact A1/A4), hence dense. Then  $q(x, -) \equiv 0$  for only finitely many  $x$ , and no vertical divisorial component exists. For the fiber: over general  $x$ ,  $C_x \cap \{q(x, -) = 0\}$  is three points by Bézout, and by Remark 3.3 the three residuals  $r_i$  lie there; that they are pairwise distinct and off  $L$  for general  $x$  is an open condition on  $F$  satisfied at an explicit witness (see Lemma 3.5 and artifact A1/A4), so the fiber is exactly  $\{r_1, r_2, r_3\}$ , reduced.

(3)  $\overline{\tau(S)} \subseteq T_q$  by (2); both have degree 3 over  $\mathbf{P}_x^2$  ( $T_q \cdot c_x = 3$  by Lemma 2.2), and by (2) the generic fiber of  $T_q$  is exactly the injective image of the generic fiber of  $S$ . Hence  $T_q$  is the closure of the image of the irreducible surface  $S$  (irreducible), and  $\tau$  is generically injective of degree 1.  $\square$

**Lemma 3.5** (Openness of the product projection). *Let  $\mathcal{U} \subset \mathbf{P}^{59} \times \mathbf{P}_x^2$  be the open locus of pairs  $(F, x)$  where  $g_x$  has distinct roots,  $q_{F(x, -)} \neq 0$ , the three residuals are pairwise distinct, and none lies on  $L$ . Then  $\{F : \exists x, (F, x) \in \mathcal{U}\}$  is open in  $\mathbf{P}^{59}$ , because the product projection  $\mathbf{P}^{59} \times \mathbf{P}_x^2 \rightarrow \mathbf{P}^{59}$  is flat and locally of finite presentation, hence open. A single explicit pair  $(F_0, x_0) \in \mathcal{U}$  therefore makes this open nonempty, hence dense. Concretely, all four conditions are checked at a witness by:  $\text{disc}(g_{x_0}) \neq 0$ ;  $q \neq 0$ ;  $\text{disc}(F(x_0, -)|_{\{q=0\}}) \neq 0$  (three distinct residuals, no root extraction needed); and  $F(x_0, -) \neq 0$  at the point  $\{q=0\} \cap L$ .  $\square$*

*Remark 3.6* (Independent Chow-theoretic check of the class). In  $\mathbf{P}_x^2 \times L_p \times \mathbf{P}_r^2$  with hyperplane classes  $H, P, R$ , the incidence  $\{p \in C_x \cap L, r \in T_p C_x \cap C_x\}$  is a complete intersection of classes  $(2H+3P)$ ,  $(2H+2P+R)$ ,  $(2H+3R)$ ; pushing down  $L_p$  gives  $20H^2 + 36HR + 9R^2$ . Subtracting twice the pushed diagonal (tangency multiplicity 2) leaves

$$20H^2 + 32HR + 3R^2 = (2H + 3R)(10H + R),$$

i.e. the residual cycle is a divisor of class  $(10, 1)$  on the  $(2, 3)$  hypersurface — the same answer by an independent route.

**3.3. The explicit parametrization.** For computations (and for the external verification of §5) it is convenient to have the composite map in closed staged form. With  $L = \{W = 0\}$ , parametrize  $L$  by  $y(t) = (t:1:0)$ , and suppose (after the harmless linear specialization described in §5, A4) that  $e = (1:0:0)$  lies on every conic  $F(-, y(t))$ . Writing  $A_{ij}(t)$  for the restricted coefficients of the conic:

$$\begin{aligned} \text{(I)} \quad x^*(t, m) = & (A_{11} + mA_{12} + m^2A_{22}, \\ & -(A_{01} + mA_{02}), -m(A_{01} + mA_{02})) \end{aligned}$$

This is the second point where the line through  $e$  in direction  $(0, 1, m)$  meets the conic. Hence it birationally parametrizes  $S$  by  $(t, m)$ .

$$\begin{aligned} \text{(II)} \quad f^* = & F(x^*(t, m), -), \quad (\alpha, \beta, \gamma) = \nabla_y f^*|_{y(t)}, \quad z^* = (\gamma, 0, -\alpha), \\ f^*(s y(t) + u z^*) = & u^2(A s + B u), \quad r(t, m) = B y(t) - A z^* \end{aligned}$$

( $z^*$  is a second point of the tangent line, since  $\alpha\gamma + \gamma(-\alpha) = 0$ ; the two leading coefficients vanish identically, expressing tangency, and  $r$  is the residual).

$$\begin{aligned} \text{(III)} \quad d = & (0, 1, n), \quad Q = F(-, r), \\ B_Q(u, v) = & Q(u + v) - Q(u) - Q(v), \\ x'(t, m, n) = & Q(d) x^* - B_Q(x^*, d) d. \end{aligned}$$

Then  $\Phi(t, m, n) = (x'(t, m, n); r(t, m))$  is a dominant rational map  $\mathbf{A}^3 \dashrightarrow X$  of degree 20. Stage (III) is the chord construction on the conic fiber through the marked point  $x^*$ , and its correctness is the one-line *reflection identity* for any ternary quadratic form  $Q$  with polarization  $B_Q$ :

$$Q(Q(d)x - B_Q(x, d)d) = Q(d)^2 Q(x),$$

so  $F(x', r) = Q(d)^2 F(x^*, r) = 0$ . The degree over  $X$  equals  $\deg(T/\mathbf{P}_y^2) = T \cdot f_y = 20$ .



*Remark 3.7* (Scope). The construction uses the plane-cubic fibration  $\rho$ , which exists precisely because the second degree is 3. It therefore says nothing about arbitrary standard conic bundles with smooth degree-9 discriminant (the matrix entries need not be all cubic), nor about bidegrees  $(2, d)$ ,  $d \geq 4$ , where the fibers of  $\rho$  have genus  $> 1$  and the tangent line meets the fiber in  $d - 2 > 1$  residual points (a correspondence, not a map). Harris's question for  $d$  large [8, p. 24] remains open.

*Remark 3.8* (Byproducts in the repository). Two independent negative results certified in the same project delimit how sharp the construction is: on a general  $X$ , *every* member of  $|H_x|$  is a  $K3$  double plane (its branch sextic never acquires a point of multiplicity  $\geq 3$ ), and *no* member of  $|H_x + H_y|$  has rational normalization. So the minimal-class searches fail provably, and the first horizontal rational surface found lives in  $|10H_x + H_y|$  — a singular member, as forced by adjunction (Lemma 2.2 ff.).

#### 4. REMOVING THE GENERALITY HYPOTHESIS

We now prove Theorem 1.1. The missing uniform point is not the resultant identity, which is universal, but horizontality: after a special specialization the three degree-10 coefficients of the residual line may have a common factor, and for a poorly chosen  $L$  the residual line could be constant in  $x$ . Varying  $L$  in the full dual plane rules this out.

**4.1. The residual-line Lattès map.** Let  $C \subset \mathbf{P}^2$  be a smooth plane cubic. We work over a field of characteristic zero. Write

$$T_p C \cdot C = 2p + g(p).$$

After choosing a flex as origin,  $g(p) = -2p$ . Thus  $g$  is a finite morphism of degree 4 and preserves collinearity. If  $L \cdot C = p + q + r$ , the three points  $g(p), g(q), g(r)$  are again collinear. They define a morphism

$$(4.1) \quad \delta_C : (\mathbf{P}^2)^\vee \longrightarrow (\mathbf{P}^2)^\vee, \quad L \longmapsto \overline{g(p)g(q)g(r)}.$$

**Lemma 4.1** (Critical values). *The morphism  $\delta_C$  has algebraic degree 4, and its reduced critical-value curve is the dual sextic  $C^\vee$ . In particular  $\delta_C$  determines the embedded cubic  $C$  by biduality.*

*Proof.* Over an algebraic closure choose a flex as origin. The morphism

$$\pi : C \times C \longrightarrow (\mathbf{P}^2)^\vee, \quad (p, q) \longmapsto \overline{pq},$$

with the tangent line as its value on the diagonal, is the quotient by the natural  $S_3$ -action on the triple  $(p, q, -p - q)$ . Its ramification divisor is the union of the three reflection curves on which two entries coincide, and its branch curve is  $C^\vee$ . Put  $\mu = [-2] \times [-2]$ . Then

$$(4.2) \quad \pi\mu = \delta_C\pi, \quad \pi^* R_{\delta_C} = \mu^* R_\pi - R_\pi = \sum_{i < j} \sum_{0 \neq \varepsilon \in C[2]} \{p_i - p_j = \varepsilon\}.$$

The curves on the right map to, and sweep out, the locus where two residual points coincide, namely  $C^\vee$ . This proves the critical-value assertion. Since  $[-2]^* \mathcal{O}_C(1)$  has degree four times that of  $\mathcal{O}_C(1)$ , the induced plane map has algebraic degree 4. Finally  $(C^\vee)^\vee = C$ .

This construction and the critical-value statement also appear explicitly in [13, p. 964, (2.6)]; see also [7, §4.3]. The quotient proof above is algebraic and therefore applies over any characteristic-zero extension field.  $\square$

**Lemma 4.2** (Constant-value descent). *Let  $K = \mathbf{C}(s, t)$  and  $\varphi : \mathbf{P}_K^n \rightarrow \mathbf{P}_K^m$  be a morphism. If  $\varphi(z) \in \mathbf{P}^m(\mathbf{C})$  for a Zariski-dense set of  $z \in \mathbf{P}^n(\mathbf{C})$ , then  $\varphi$  is defined over  $\mathbf{C}$ .*

*Proof.* Pass to source and target affine charts retaining a dense subset. Write the coordinate ratios as  $R_i(z) \in K(z_1, \dots, z_n)$ . Applying either  $\mathbf{C}$ -derivation of  $K$  shows that  $D(R_i)$  vanishes at a dense set of constant points, hence vanishes identically. The common constants of these derivations in  $K(z_1, \dots, z_n)$  are  $\mathbf{C}(z_1, \dots, z_n)$ , so all ratios descend. The descended coordinate forms have no common zero because their base change is the given morphism.  $\square$

**4.2. Choosing a horizontal line section.** Let  $X$  be an arbitrary smooth  $(2, 3)$  hypersurface and put

$$K = \mathbf{C}(\mathbf{P}_x^2), \quad C = X_K \subset \mathbf{P}_{y,K}^2.$$

The generic cubic  $C$  is smooth by generic smoothness. For completeness, smoothness of  $X$  rules out a fiber equal to the whole  $\mathbf{P}_y^2$ : at such a fiber the two derivatives in independent local  $x$ -directions are plane cubics and have a common zero, which would be singular on  $X$ .

The embedded cubic  $C$  is not constant. Otherwise its equation over  $K$  would be  $\lambda(x)f_0(y)$  for a fixed cubic  $f_0$ . Comparing one nonzero coefficient makes  $\lambda$  a quadratic form, so  $F(x, y) = Q(x)f_0(y)$  in  $\mathbf{C}[x, y]$ , contradicting smoothness.

Apply Lemma 4.1 to  $C/K$ . The morphism  $\delta_C$  cannot be defined over  $\mathbf{C}$ : if it were, its critical-value sextic and then its bidual  $C$  would be constant. By Lemma 4.2, the constant lines having constant image under  $\delta_C$  are not Zariski dense. We may therefore choose a constant line  $L \subset \mathbf{P}_y^2$  such that

$$(4.3) \quad \delta_C(L) \notin (\mathbf{P}_y^2)^\vee(\mathbf{C}),$$

and simultaneously impose the following open conditions:  $L$  is not contained in the conic discriminant,  $C \cap L$  is reduced, and  $[-2]$  is injective on its three points. The last bad locus is, after algebraic closure, the finite union of curves swept out by the chords  $\overline{p, p + \varepsilon}$  for  $0 \neq \varepsilon \in C[2]$ ; the union is Galois invariant. A proper closed subset of the constant dual plane base-changed to  $K$ , together with these proper  $K$ -closed loci, cannot contain every constant point (expand its finitely many  $K$ -coefficients in a finite  $\mathbf{C}$ -linearly independent set). Hence such an  $L$  exists.

The surface

$$S_L = X \cap (\mathbf{P}_x^2 \times L)$$

is integral: its generic fiber over  $L$  is a geometrically integral conic, and smoothness of  $X$  also rules out a whole  $\mathbf{P}_x^2$  fiber by the same two-derivative argument. Tsen's theorem then gives  $\mathbf{C}(S_L) = \mathbf{C}(t)(z)$ , so  $S_L$  is rational. Tangent residuals define  $\tau_L : S_L \dashrightarrow X$ . The chosen injectivity condition shows that its image  $T_L$  is irreducible,  $S_L \dashrightarrow T_L$  is birational, and  $T_L \rightarrow \mathbf{P}_x^2$  has degree 3.

Suppose  $T_L$  were not horizontal over  $\mathbf{P}_y^2$ . Its image cannot be a point, since every conic fiber is one-dimensional. If its image were a curve, write  $[T_L] = aH_x + bH_y$  using Lemma 2.2. A general conic fiber outside that curve is disjoint from  $T_L$ , so  $2a = 0$ ; degree 3 over  $\mathbf{P}_x^2$  gives  $3b = 3$ . Thus  $[T_L] = H_y$ . The restriction sequence gives

$$H^0(X, \mathcal{O}_X(0, 1)) = H^0(\mathbf{P}_y^2, \mathcal{O}(1)),$$

so  $T_L$  would be the inverse image of a constant line  $M$ . On the generic cubic this says  $\delta_C(L) = M$ , contradicting (4.3). Hence  $T_L$  is horizontal.

Finally put  $L = \{W = 0\}$  and use Theorem 3.2. The residual line has three degree-10 coefficients in  $x$ . After dividing their gcd of degree  $e$ , its primitive equation cuts a divisor

$$D \sim (10 - e)H_x + H_y$$



containing  $T_L$  as its unique component dominating  $\mathbf{P}_x^2$ . We have  $[T_L] = aH_x + H_y$ . Every effective prime divisor  $AH_x + BH_y$  has  $A, B \geq 0$ , by intersection with general conic and cubic fibers. Comparing coefficients in  $D$  therefore gives

$$1 \leq a \leq 10 - e \leq 10.$$

The normalization of  $T_L$  is rational and its degree over  $\mathbf{P}_y^2$  is  $2a \leq 20$ . Base change of the conic bundle to this normalization has the tautological point and is birational to  $\tilde{T}_L \times \mathbf{P}^1$ . Projection back to  $X$  proves Theorem 1.1.

*Remark 4.3* (A broader rational case). If an integral (2, 3) member has geometrically integral but singular generic cubic over  $\mathbf{C}(\mathbf{P}_x^2)$ , its unique geometric singular point is rational over that field. Projection from it makes the generic cubic rational, so  $\mathbf{C}(X) = \mathbf{C}(\mathbf{P}_x^2)(t)$  and  $X$  itself is rational. Thus the smooth generic cubic is the hard case. Generically geometrically reducible or nonreduced cubic fibrations require separate arguments.

## 5. VERIFICATION ARTIFACTS

The all-smooth proof uses one machine-checked algebraic fact, the universal identity (Theorem 3.2); its remaining inputs are the prose Lattès, descent, and Picard arguments of Section 4. The sharper generic degree-20 statement additionally uses primitivity of  $q$  and the explicit witnesses of Lemma 3.5. These finite facts are machine-checked. This section is a complete ledger. All paths are relative to the repository <https://github.com/mattrobball/unirational>; logs of every run are committed beside the scripts.

**A0 — the all-smooth certificate.** The complete standalone proof, including the quotient ramification formula, constant-value descent, simultaneous choice of  $L$ , and degree bound, is `certificates/all_smooth_tangent_residual_theorem.md`.

**A1 — the certificate of record.**

`problems/B-conic-bundle-multisections/certificates/tangent_residual_local_checks.py` (with `.log`):

- verifies (2) *fully symbolically* over the universal ring  $\mathbf{Z}[a_3, \dots, k]$  in ten independent coefficient symbols (the  $W^0, W^1$  coefficients of  $\mathcal{R} + \Delta f$  vanish identically; the quotient by  $W^2$  is linear);
- verifies each coefficient of  $q_f$  is homogeneous of degree 5 and that the three have gcd 1 universally, and again after a recorded substitution by ten quadratic forms (so the general (2, 3) equation has primitive  $q$ );
- supplies the smooth-fiber witness of Lemma 3.5: an explicit  $(F_0, x_0)$  with the cubic smooth (unit Gröbner bases in three charts), three residual points computed, verified to lie on the cubic, on the tangent lines, on  $\{q = 0\}$ , off  $L$ , pairwise distinct;
- records the divisor arithmetic  $(10, 3) - 2(0, 1) = (10, 1)$ , degree 20.

The full prose proof of the generic refinement is `certificates/tangent_residual_theorem.md`.

**A2 — independent re-derivation (audit).** During the audit of this session, the identity was re-derived from scratch in a fresh SymPy session sharing no code with A1: the  $W^0$  divisibility, the cofactor being the *classical* discriminant (both sign conventions tested), degree-5 homogeneity, and universal gcd = 1 were reconfirmed independently.

**A3 — independent end-to-end geometric check (audit).** On an explicit cubic with rational points on  $L$  ( $g = (u - v)(u - 2v)(u + v)$ , generic  $h, i, j, k$ ), the three tangent residuals were computed *from the raw geometry* (tangent lines; factoring the double root out of the restricted cubic) and verified to be distinct, off  $L$ , and to lie on  $\{q_f = 0\}$  — confirming the geometric interpretation, not merely the algebra. A second check drew a random  $(2, 3)$  form over  $\mathbf{Q}$  and verified the witness conditions of Lemma 3.5 at  $x_0 = (1, 2, 3)$ , plus  $\gcd(q_U, q_V, q_W) = 1$  by restriction to two lines.

**A4 — external random-coefficient instance.**

`problems/B-conic-bundle-multisections/external_check/`: a full end-to-end falsification test on coefficients drawn from `/dev/urandom` (recorded), with four coefficients zeroed so that the section  $e = (1:0:0)$  exists over  $\mathbf{Q}$  and the composite map has exact rational coefficients (a codimension-4 linear slice needed only because Tsen’s theorem is not effective over  $\mathbf{Q}$ ; over  $\mathbf{C}$  the theorem needs no such conditioning). The harness `build_explicit_map.py` constructs the staged map of §3.3 and verifies, exactly:

| check | statement                                         | method                                 |
|-------|---------------------------------------------------|----------------------------------------|
| C1    | $F(x^*, y(t)) \equiv 0$                           | symbolic                               |
| C2    | tangency (two leading coefficients vanish)        | symbolic                               |
| C3    | $f^*(r) \equiv 0$                                 | full $100 \times 53$ grid <sup>†</sup> |
| C4    | $q_{f^*}(r) \equiv 0$                             | full $62 \times 37$ grid <sup>†</sup>  |
| C5    | reflection identity                               | generic symbols                        |
| C6    | $F(\Phi) = 0$ at sample rational points           | exact evaluation                       |
| C7    | Jacobian rank 3 (dominance)                       | exact symbolic differentiation         |
| C8    | fiber over random $y_0$ : squarefree of degree 20 | resultant                              |

<sup>†</sup>Deterministic proofs: a polynomial of bidegree at most  $(D_1, D_2)$  vanishing at a  $(D_1+1) \times (D_2+1)$  grid of distinct rational nodes is zero; the bidegree bounds  $(99, 52)$  and  $(61, 36)$  are derived in the harness.

All eight pass (`run_log.txt`); the explicit map instance is `map_instance.txt`, and `README_CHECK.md` gives CAS-agnostic instructions for third-party verification with no trust in the pipeline.

**A5 — unconditioned instance (no slice).** `external_check/unconditioned_check.py` (with `.log`) addresses the natural objection that the codimension-4 slice in A4 might matter: it runs the full mechanism on a fresh entropy draw with *no* coefficient zeroed. With no rational point available, it works in the cubic étale algebra  $K = \mathbf{Q}[\theta]/(g_0)$ ,  $g_0 = F(x_0, -)|_L$ , where the three marked points live; reduction modulo the full cubic verifies each identity at all three conjugate points simultaneously. It checks: primitivity of  $q_F$ ; the witness conditions of Lemma 3.5; tangency,  $F(x_0, r) = 0$ ,  $q(r) = 0$ , and  $r \notin L$  (via  $\text{Res}(g_0, r_W) \neq 0$ ) in  $K$ ; the stage-(III) point  $F(x', r) = 0$  in  $K$ ; and the squarefree degree-20 fiber. All pass. The slice in A4 is thereby confirmed to play no role beyond supplying  $\mathbf{Q}$ -rationality of the composite parametrization (Tsen’s theorem being non-effective over  $\mathbf{Q}$ ); by  $\text{PGL}_3 \times \text{PGL}_3$ -equivariance the sliced family  $\{X \supseteq \{e\} \times L\}$  is in no special position for anything the proof uses.

**Prose-only steps.** For completeness: the steps of the proof resting on standard mathematics rather than computation are Tsen’s theorem, generic smoothness, Grothendieck–Lefschetz (Lemma 2.2), the group-law and Lattès quotient arguments, constant-value descent, openness of the flat product projection in Lemma 3.5, and the Chow-theoretic cross-check. Each is supplied in the proof or cited to a primary source.

**Independence and audit trail.** A1 was rerun in a clean environment during the audit (Macaulay2 1.26.06 / SymPy 1.14.0) with output matching the committed logs; A2–A3 share no code with A1; A4 takes its input from an external entropy source. The literature verification was performed against primary sources on 2026-07-22 and re-confirmed on 2026-07-23 by reading the relevant papers directly. Specifically checked: [10, Corollary 4.9] and its definition  $\delta := \deg D$ , which places the published boundary for a *smooth* discriminant at  $\delta \leq 8$  and leaves our  $\delta = 9$  uncovered by exactly one unit; [10, Corollary 4.10] ( $\delta \leq 12$  for quadric surface bundles over  $\mathbf{P}^2$ , which subsumes the  $\mathbf{P}^3 \times \mathbf{P}^2$  analogue of Theorem 1.1); [10, Remark 2.9] for Lang’s theorem in the form  $h > 2^{r+n-h} - 2$ ; [3, Theorem 1.1]; [12, Corollary 1.2]; the method and status of [6]; and the residual-line Lattès statements in [13, 7]. One caution recorded for future searches: the bidegree- $(2, 3)$  hypersurfaces in  $\mathbf{P}^2 \times \mathbf{P}^3$  of Auel–Böhning–Pirutka are *cubic surface bundles* over  $\mathbf{P}^2$ , a different family from the one studied here.

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